

SENIOR SOFTWARE ENGINEER · SYSTEMS &
PLATFORM ARCHITECTURE · APPLIED AI

EMMANUEL MOBOLAJI ARO

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REMOTE / RELOCATION

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// EXECUTIVE PROFILE

I don't just build features — I **architect systems**. Senior Full Stack Software Engineer with **6+ years** shipping distributed platforms, Agentic AI pipelines, and production-grade infrastructure across **7 companies** and **4 continents**. Master of the Python (Django / Django Ninja) and TypeScript (Next.js App Router) ecosystems. Currently engineering legal AI and GPU processing infrastructure at **LunarTech (Austin, TX)** while independently shipping **Geod AI** — a sovereign, multi-model AI workspace. I care about clean domain-driven design, defensible API contracts, and systems that hold under load.

// TECHNICAL EXPERTISE

CORE MASTERY ★

Python, JavaScript, TypeScript, PHP, Laravel, Django Ninja, Next.js (App Router), React, PostgreSQL, MySQL, Docker, REST APIs, JWT/OAuth2

AI / LLM STACK

Agentic RAG (LlamaIndex / LangGraph / LangChain), Ollama (Local LLMs), OpenRouter, Gemini AI, WhisperX, NLP Pipelines, OpenCV, Scikit-learn, TensorFlow

BACKEND & CLOUD ARCH.

Microservices, Event-Driven Systems, GraphQL, Node.js, AWS (EC2, S3, RDS, IAM), Modal (GPU serverless), Backblaze B2, Nginx

DATA & DEVOPS

MySQL, SQLite, Supabase, Redis, Schema Modeling, Query Optimisation, CI/CD (GitHub Actions), Linux, Postman, Git

ENGINEERING LEADERSHIP

System Architecture Design, API Contract Design, Security Hardening (RBAC, JWT), Agile / Scrum, Code Review & Mentorship

LunarTech

CURRENT

May 2026 - Present | Austin, TX, US (Remote)

Software Engineer — AI Systems & Platform Engineering (Apprenticeship)

IMPACT: Shipped Rosendahl (Legal AI) & Dark Phoenix (GPU pipeline) end-to-end; QA'd Octavia — database optimisation recommendations adopted team-wide to support thousands of concurrent users without degradation.

- Contributing to **Rosendahl**, LunarTech's flagship legal AI platform — a self-hosted agentic workspace built as “Claude Code for law” — engineering Agentic RAG pipelines (LlamaIndex / LangGraph) across 121+ active ECHR case workspaces with fully evidence-linked, formally verified outputs.
- Designing legal document processing APIs using LexNLP and Blackstone (spaCy) for automated named-entity extraction, citation parsing, and admissibility-risk scoring; enforcing a four-tier confidential AI model routing strategy (OpenRouter / Claude / Ollama local) ensuring privileged legal data never exits the local inference lane.
- Delivered **Dark Phoenix**, LunarTech's AI podcast video clipper — architecting the end-to-end GPU pipeline on Modal (TalkNet → WhisperX → Gemini moment selection → ffmpeg clip render), YouTube ingestion via yt-dlp, provider-agnostic S3 storage (Backblaze B2), and branded watermark pipeline.
- Conducted comprehensive QA testing and performance analysis on **Octavia** — LunarTech's world-class AI dubbing platform that translates video, audio, and subtitles into 60+ languages with 99% lip-sync accuracy and 10ms latency via advanced voice cloning (10,000+ hours translated). Identified critical database bottlenecks under multi-user concurrency load and delivered optimisation recommendations formally adopted by the engineering team — enabling the platform to scale for thousands of simultaneous users processing real-time translations without degradation.
- Resolved 8 critical production deployment failures spanning Modal GPU image builds, WhisperX/setuptools dependency conflicts, and Next.js RSC compiler violations — bringing the platform from broken to fully deployed on Vercel and Modal.

Boxonia Blueprint (boxonia.com)

CURRENT

Oct 2024 - Present | Lagos, NG

Software Engineer (Platform & Ecosystem Architecture)

IMPACT: Built Boxonia's entire digital presence from the ground up — single-handedly brought a Nigerian 360° film production & talent management studio fully online, from zero infrastructure to a live, production-grade platform.

- Conceived, designed, and engineered Boxonia's complete digital platform from zero — the company had no web presence before this engagement. Single-handedly brought a Nigerian 360° film production & talent management studio (boxonia.com) fully online, delivering a production-grade platform for story development, talent management, and project showcasing across a globally distributed creative operation.
- Currently architecting and maintaining the full platform ecosystem in Python — story development workflow engine, talent directory, project portfolio showcase, and Spotlight section — purpose-built for African storytelling at scale across film, TV, and commercials, consolidating workflows across 5+ distributed production teams.
- Continuing to evolve the talent-booking and cross-border coordination system — enabling zero-friction client onboarding and operating at an international production cadence with a 40% reduction in scheduling overhead.
- Maintaining and scaling relational schemas and secure RESTful APIs, ensuring high-availability data synchronisation between production units under strict SLA requirements as the company's audience and production volume grows.

Ornament & Crime (ornamentandcrime.com)

Aug 2024 - Mar 2026 | Birmingham, AL, US

Frontend Engineer (Shopify Developer)

IMPACT: Rebuilt the entire Shopify Plus storefront from the ground up — 48% faster load time, 31% checkout conversion uplift, zero downtime. The kind of migration most agencies take months to plan; delivered in weeks.

- Architected a full Shopify Plus storefront rebuild from legacy theme to a componentised Liquid 2.0 + Tailwind CSS system, slashing page load time by 48% and directly attributing to a 31% uplift in checkout conversion over the following quarter.
- Engineered custom Shopify app integrations (inventory sync, loyalty tiers, fulfilment webhooks) via the Shopify Admin GraphQL API, eliminating an entire category of manual operations and saving 20+ staff hours per week.
- Led the brand's migration to headless-compatible architecture, setting the engineering foundation for future Next.js storefront decoupling without a single day of downtime.

Tech4mation (NativeTalk) (nativetalk.io)

Feb 2026 - Jun 2026 | Lagos, NG

Senior Software Developer, Platform Engineering

IMPACT: Shipped NativeTalk CRM (nativetalk.io/crm) end-to-end across frontend and backend — reset company delivery velocity, resolved critical customer-facing issues faster than any prior engineering cycle, and formally awarded by leadership for measurable impact on customer satisfaction.

- Led full-stack engineering on **NativeTalk CRM** — an enterprise-grade omnichannel command center (tasks, tickets, leads, AI reporting) serving large businesses and multiple organisations — working directly with Product Managers to ship features at a velocity that redefined the company's engineering benchmark.
- Designed and implemented end-to-end CRM features (ticket management, lead pipeline views, AI reporting dashboards) using Django Ninja on the backend and Next.js App Router on the frontend, ensuring every release shipped clean, tested, and defect-free.
- Drove aggressive bug-triage cycles against live customer SLA commitments — resolving critical production issues at a pace that directly improved customer satisfaction scores and earned formal recognition and reward from company leadership.
- Architected **TaskForge Pro**, a high-throughput internal team-management platform (Django Ninja · Next.js App Router · PostgreSQL) with decoupled microservice boundaries and strict typed API contracts.
- Standardised engineering environments via containerised Docker workflows, cutting new-engineer onboarding from 3 days to under 2 hours.

Bincom Dev Center

Sep 2025 - Jan 2026 | Lagos, NG

Full Stack Developer

IMPACT: Drove frontend performance gains (35% faster FCP) and database optimizations while mentoring the next wave of tech talent.

- Joined Bincom Dev Center — an incubator and catalyst for tech excellence — to drive the development of innovative tech solutions while simultaneously nurturing the next generation of engineering talent.
- Architected high-performance, SEO-optimized frontend interfaces using Next.js (App Router) and React. Leaned heavily into Server Components and smart data-fetching patterns to slash First Contentful Paint (FCP) by 35%.
- Got my hands dirty in the database layer: overhauled relational MySQL schemas and refactored heavily-used SQL queries to eliminate latency spikes and speed up complex data retrieval by 20% under peak load.
- Took on a mentorship role, guiding junior developers through the trenches of TypeScript best practices, modular component design, and solid Git workflows to elevate the whole team's delivery velocity.

CyberBuddies (cyberbuddies.com)

Apr 2024 - Aug 2024 | Ibadan, NG

Frontend Engineer

IMPACT: Shipped production-ready frontend pages for an AI-powered Business Operating System™ at an early product stage — UI patterns and component architecture established became the foundation of the product's visual language.

- Joined **CyberBuddies** — an AI-powered Business Operating System™ for SMBs that deploys AI business partners (AI CFO, AI COO, AI Executive Team™) to help business leaders automate operations, understand performance, and make better decisions — as a Frontend Engineer at a pivotal early stage of the product.
- Built and iterated on high-quality frontend UI pages and components, translating complex AI product concepts into clean, intuitive, and accessible interfaces using modern frontend practices; UI patterns established during this period became the foundation of the platform's visual language.
- Contributed meaningfully to the team's delivery velocity — shipping production-ready code, participating in code reviews, and raising the bar on frontend standards at an early-stage company where every engineer's output has outsized impact.
- Supported migration of production workloads to AWS infrastructure, applying optimisation strategies that reduced cloud operating costs by 18% while maintaining 99.9% uptime SLA targets.

SafeSpace.org Association

May 2023 - Feb 2024 | California, US (Remote)

Operations QA & Automation Engineer

IMPACT: Automated review pipelines in Python/Django — 30% throughput gain, 50% fewer data errors.

- Deployed Python/Django automated review pipelines replacing manual verification loops — 30% throughput gain, 50% fewer data-entry errors.
- Instituted rigorous automated testing protocols in containerised Docker environments, establishing company-wide benchmarks for system integrity.

Fiatci.io

Sep 2022 - Mar 2023 | Brussels, BE (Remote)

Lead Full Stack Engineer - WHERE IT ALL BEGAN

- IMPACT:** The humble beginning — joined as a young developer and within months, through relentless passion, obsession with the craft, and an unshakeable work ethic, was hand-picked to lead a 7-engineer team. Shipped the company's first ever successful commercial product launch.
- **Where it all began.** Joined Fiatci.io early in my career as a young developer — and within just a few months, driven by an obsession with the craft, relentless hard work, and a passion that made me outpace engineers with years more experience, was chosen by the company to lead the engineering team. That single moment at a young age ignited everything: it proved that raw talent and dedication speak louder than years on a résumé.
 - Stepped up to the Lead Full Stack Engineer role at a formative stage of my career — taking full ownership of a 7-person cross-functional engineering team, driving the end-to-end architecture, development, and production deployment of the company's flagship platform, and delivering the **first successful commercial product launch in company history**.
 - Owned full technical direction across frontend, backend, and DevOps infrastructure — establishing rigorous code review standards, CI/CD release strategies, and zero-downtime production cutover protocols that set the engineering baseline for the entire team.
 - This early leadership opportunity — earned, not given — built the confidence, accountability, and engineering instinct that define how I operate today as a global talent.

// FLAGSHIP SYSTEMS & AI PROJECTS

Geod — SOVEREIGN AI WORKSPACE <code>NEXT.JS</code> · <code>LLMS</code> · <code>WEBRTC</code> · <code>REDIS</code> Architected and shipped a unified real-time collaboration + AI workspace — four proprietary AI engines, sandboxed code runtime, multi-modal OCR pipeline, and zero vendor lock-in by design. geod-psi.vercel.app	Rosendahl — LEGAL AI PLATFORM <code>LANGGRAPH</code> · <code>LLAMAINDEX</code> · <code>OLLAMA</code> Agentic RAG pipelines + legal document processing APIs (LexNLP / Blackstone) for ECHR case structuring across 121+ active workspaces with four-tier confidential model routing.
Dark Phoenix — GPU PODCAST CLIPPER <code>MODAL</code> · <code>WHISPERX</code> · <code>GEMINI AI</code> · <code>FFMPEG</code> End-to-end AI video pipeline: TalkNet speaker detection → WhisperX transcription → Gemini moment selection → branded MP4 export. Deployed on Modal + Vercel.	Nativetalk — SUPPORT DASHBOARD <code>NEXT.JS</code> · <code>TYPESCRIPT</code> · <code>REST APIS</code> Architected a centralized ticket-management dashboard for enterprise support — tracking inquiries, managing agent assignment, and surfacing real-time completion-rate metrics via strict typed API contracts.
Enterprise Omnichannel CRM <code>NEXT.JS</code> · <code>DJANGO NINJA</code> · <code>POSTGRESQL</code> Led full-stack engineering on an enterprise command center handling tasks, tickets, leads, and AI reporting. Shipped high-impact features bridging frontend UI with decoupled microservice boundaries.	Octavia — AI DUBBING PLATFORM <code>VOICE CLONING</code> · <code>PERFORMANCE QA</code> Performance analysis and QA for an AI dubbing platform translating media into 60+ languages with 99% lip-sync accuracy. Resolved database bottlenecks to scale for thousands of simultaneous users.

Enterprise Malware Detection Engine PYTHON · SCIKIT-LEARN · SECURITY ML

Supervised ML platform for real-time endpoint malware classification with automated feature extraction and continuous retraining pipelines for evolving threat corpora.

Clinical Oncology Risk Model SCIKIT-LEARN · PANDAS · XAI

Explainable diagnostic classification model trained on clinical datasets. Prioritised interpretable feature engineering for clinical decision support.

Computer Vision Attendance Platform OPENCV · PHP · MYSQL

High-speed biometric attendance system integrating real-time webcam capture with OpenCV computer-vision pipelines and normalisation against a secured relational store.

// EDUCATION & CERTIFICATIONS

B.Sc. in Computer Science Federal University Oye-Ekiti, Nigeria	2025
Full Stack Engineer Certificate Google (Google Developer Student)	2024
Advanced Software Engineering Certification GB-Tech Learning Centre, Nigeria	2023